



SQL Server Encryption and Encryption Types



Encryption

- Data protection is critical for ensuring that your organization is compliant with regulatory compliance standards like the GDPR and for meeting the expectations of your clients and business partners.
- Not only can data breaches result in large fines, but the reputational damage can be just as great.
- Encryption is the process of obfuscating data by the use of a key or password.
- This can make the data useless without the corresponding decryption key or password.
- Encryption does not solve access control problems. However, it enhances security by limiting data loss even if access controls are bypassed.
- For example, if the database host computer is misconfigured and a hacker obtains sensitive data, that stolen information might be useless if it is encrypted.



Encryption Types

- **SSL Transport Encryption :**
- Like websites that secure traffic between browser and server, SQL Server can be configured to use Secure Sockets Layer (SSL) to encrypt traffic as it travels between the server instance and client application.
- Additionally, the client can validate the server's identity using the server's certificate.
- SSL only protects data as it travels across the network, but, unlike most of the other forms of SQL Server encryption, SSL is available in all supported versions of SQL Server and in all editions.
- Before enabling SSL, you'll need to install a certificate on the SQL Server. The best way to do this is by requesting a certificate from your own enterprise certification authority (CA).



Encryption Types

- **SQL Server Transparent Data Encryption (TDE) :**
- Transparent Data Encryption (TDE) in SQL Server protects data at rest by encrypting database data and log files on disk.
- It works transparently to client existing applications, so they don't need to be changed when TDE is enabled.
- TDE uses real-time encryption at the page level.
- Pages are encrypted before they are written to disk, without increasing the size of your data and log files, and pages are decrypted when read into memory.
- TDE is available only in Enterprise editions of SQL Server.



Encryption Types

- **Column/Cell-Level Encryption :**
- Available in all editions of SQL Server, cell-level encryption can be enabled on columns that contain sensitive data.
- The data is encrypted on disk and remains encrypted in memory until the DECRYPTBYKEY function is used to decrypt it.
- Therefore, although the SQL data is encrypted, it is not secure beyond simply using a function in the user context to decrypt it.
- Additionally, because a function is needed to decrypt the data, client applications must be modified to work with cell-level encryption.



Encryption Types

- **Backup Encryption :**
- Backup Encryption works like TDE but encrypts SQL backups instead of the active data and log files.
- Backup Encryption is available in SQL Server 2014 and later.
- You can specify AES 128, AES 192, AES 256 or Triple DES encryption, and use either a certificate or asymmetric key stored in EKM.
- Additionally, it is possible to enable TDE and Backup Encryption simultaneously, although you should use different certificates or keys.
- Just as with TDE, if you enable Backup Encryption, you must also back up the certificate or key. Without the key or certificate, the backup file can't be used to restore data.
- You must have the original certificate when restoring data.



Encryption Types

- **Always Encrypted :**
- Always Encrypted encrypts sensitive data in client applications without revealing the encryption keys to the database engine,
- providing separation between data owners and data managers.
- For example, with Always Encrypted enabled, you can be sure that your database administrators won't be able to read sensitive data. As the name suggests, data is encrypted at rest.
- Always Encrypted can be configured for individual database columns.
- The encryption process is transparent to client applications but requires a special driver on client computers.
- Always Encrypted is available in SQL Server 2016 and later, but only in Enterprise editions.